

Sujay Seeram

Cyber-Physical Systems, Sovereign AI Infrastructure & Embedded Electronics

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ENGINEERING PROFILE & CORE PHILOSOPHY

Undergraduate systems and electronics engineer at Amrita Vishwa Vidyapeetham with an established track record in **cyber-physical systems**, **embedded hardware architectures**, and **physics-informed machine learning**. An open, execution-oriented learner and quick adapter driven by the “**Extreme Economic Engineering**” paradigm: deriving minimalist hardware and algorithmic configurations from physical first principles to drastically cut component costs, compute overhead, and power consumption without sacrificing capability. Beyond core engineering, deeply committed to applying technological innovation toward **renewable energy**, **environmental sustainability**, and **social responsibility**. Disciplined Naval NCC cadet and 3-time Services Selection Board (SSB) attendee.

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
– **Core Focus:** Cyber-Physical Architectures, Embedded Edge ML, High-Throughput Runtimes, Physics-Informed Modeling.
- **Navy Children School (NCS), CBSE** Visakhapatnam, India
Senior Secondary (Class XII): 92.5% | Secondary School (Class X): 95.0% *2021 – 2023*

TECHNICAL SKILLS

- **Hardware, MCUs & Protocols:** STM32 (ARM Cortex-M4), ESP32, Raspberry Pi, RP2040, SPI, I2C, UART, CAN, DMA synthesis, ADC/DAC.
- **Programming Languages:** Python, Rust, C/C++17, TypeScript, JavaScript, Go, Dart (Flutter), SQL, POSIX Shell/Bash.
- **Edge AI, DSP & Machine Learning:** PyTorch, Scikit-Learn, TinyML, Fast Fourier Transform (FFT) DSP, LanceDB, Model Context Protocol (MCP).
- **Systems, Runtimes & Architecture:** Tokio async, Actix-Web, FastAPI, FreeRTOS, Linux eBPF, Docker, WebSockets, Linux IPC.
- **Sensors & Prototyping:** ACS712 current, ZMPT101B voltage, MPU6050 6-DoF IMU, OPA1612 analog filters, KiCad PCB schematic/layout.
- **Communication & Additional Skills:** Confident Public Speaker; delivered multiple technical club workshops; proficient in **Basic Sign Language (ISL)**.

ACHIEVEMENTS, DISTINCTIONS & MILITARY DISCIPLINE

- **Rapid Prototyping Portfolio & Engineering Track Record** *Portfolio Metrics*
 - **95 Projects Explored, 50 Executed, 3 Major Hackathons Won:** Interdisciplinary distribution of **15 AI+CS+ECE** cyber-physical projects, **25** dedicated ECE/embedded systems, **5** core CS/systems software, and **24** foundational sandbox builds.
 - **Real-Time Production Deployment (SAFE):** Engineered **SAFE**, an operational warehouse asset tracking ecosystem featuring low-power BLE duty cycling and high-speed batch scanning (*details reserved for interview discussion*).
- **Major Competitions, Global Leaderboards & Distinctions** *Competitive Honors*
 - **IIT Madras Smart India Challenge:** Awarded **2nd Runner Up**; honored as **Youngest Panelist** at the summit; conducted multiple technical workshops across college clubs on embedded systems and AI.
 - **Google Virtual PromptWars Global Leaderboard:** Ranked in the **Top 400 globally across 3 separate challenge cycles** in Google for Developers × Hack2skill PromptWars, competing against thousands of builders in rapid AI prototyping.
 - **HSBC Global Train Green Program (2022):** Awarded **Best Junior Sustainability Champion** for environmental innovation.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - **NCC Senior Division:** Trained cadet in **3(A) Naval Unit NCC, Visakhapatnam (VSP)**; developed crisis leadership, nautical discipline, and obstacle course proficiency.
 - **Services Selection Board (SSB):** Attended 3 SSB selection boards (**Screened-in twice & Conferenced-out twice**; Screened-out once); demonstrated resilient situational decision-making and psychological adaptability under high-stress evaluations.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected technical delegate at **GARC Chennai Meet**; engaged with engineers on vehicle sensor failure preemption and telemetry standards.

FEATURED ENGINEERING PROJECTS (ORDERED PORTFOLIO)

I. Interdisciplinary Cyber-Physical AI (AI, CS & EC Combined)

- **1. EVolvAI – Generative AI & Physics-Constrained EV Grid Optimization** *Release v1.0.0*
 - Architected a scalable cyber-physical demand framework in PyTorch coupling a **Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm to simulate charging loads under distribution grid bus limits and climate stress anomalies.
 - Achieved $R^2 \approx 0.5$ predictive accuracy on sparse field data; built an interactive Streamlit simulation console for municipal engineers.
- **2. MotorSafe – Tri-Modal Edge-AI Predictive Maintenance Ecosystem** *Release v0.01*
 - Constructed an edge-native telemetry pipeline fusing 3 synchronized sensor channels: **ACS712** current, **ZMPT101B** voltage, and **MPU6050** vibration.
 - Engineered on-chip FFT DSP to extract spectral sidebands and deployed an unsupervised **Isolation Forest** on ESP32 to preempt bearing breakdown without cloud latency, streaming live diagnostics to an interactive Three.js 3D digital twin dashboard.
- **3. SpecRAG – Zero-Hallucination Firmware & Datasheet Auditing Gateway** *Release v0.1.0*
 - Engineered an enterprise developer tool bridging 1,000-page silicon datasheets with embedded C/C++ firmware using the **Model Context Protocol (MCP)**.
 - Performed localized vector retrieval against indexed silicon register banks to preemptively catch peripheral pin collisions, clock-tree prescaler errors, and timing hazards directly in VS Code before compilation.

II. Embedded Systems & Electronics (ECE)

- **4. AQUAPULSE – Cognitive Software-Defined Sonar Payload for AUVs** *Repository*
 - Engineered a cognitive software-defined sonar payload for Autonomous Underwater Vehicles (AUVs) meeting the SIH26058 MoES mandate.
 - Implemented zero-CPU DMA-to-DAC acoustic synthesis on **STM32 / ESP32-S3** with **OPA1612** active filtering, deploying an INT8 **TinyML** engine to dynamically adapt ping frequency (100–480 kHz) across shifting thermoclines, slashing acoustic power consumption by **38%**.
- **5. Pheno-Speak – Edge Acoustic Vocal Biomarker Extraction Engine** *Repository*
 - Built an edge audio digital signal processing pipeline for non-invasive vocal biomarker extraction aligned with SDG 3.4 (Digital Phenotyping).
 - Extracted MFCCs, jitter, and shimmer features on lightweight microcontroller hardware without offloading unencrypted audio to external endpoints.
- **6. Dust-Vision & Dust-Sim – Edge Particulate Tracking & Dispersion Engine** *Release v0.1*
 - Deployed computer vision models on edge camera modules to quantify real-time construction site particulate pollution and dust plumes.
 - Coupled sensor data with a custom particle kinematics physics simulation (**dust-sim v1.0.0**) modeling dispersion under thermal and wind shifts.

III. Systems Software, Developer Tools & Computer Science (CS)

- **7. ytstr – High-Performance Psychoacoustic Audio Engine** *Release v2.0.0*
 - Developed a terminal-native streaming runtime powered by a psychoacoustic **DecisionEngine** executing real-time spectral analysis.
 - Implemented 8 broadcast-grade transition algorithms and built an active backpressure memory pool, slashing runtime RAM by **70%** (< 260MB).
- **8. Informed-Poll – Swipeable Civic Intelligence Platform** *Release v1.0.0*
 - Developed a mobile-first civic intelligence companion featured in the Google Solution Challenge, pairing swipe mechanics with verified candidate records.
 - Integrated localized neural RAG using **LanceDB** and Gemini 3.1 Flash Lite over a Cloud Run FastAPI backend with zero-hallucination verification.
- **9. Kavach-UPI – Android Accessibility Security Shield** *Repository*
 - Authored an on-device security shield leveraging Android Accessibility APIs to detect active screen-sharing malware during financial UPI transactions.
 - Instantly deployed an opaque system window overlay to block remote touch injections while triggering localized security alerts.